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Assessment of Drought Impacts on Remotely Sensed Seasonal Snow Depletion Patterning: A Case Study Over the Boise River Basin, Idaho

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ABSTRACT

Snow is an essential source of freshwater, and remotely sensed snow cover can offer daily spatial data critical to manage and model snowmelt runoff. Cloud cover obscures daily optical remotely sensed snow cover, and uncertainty associated with cloud gap filling methods may be exacerbated by drought thereby limiting effective implementation of snow cover data into snowmelt runoff models. The goal of this research is to provide a cloud free, reliable, and dynamic estimate of daily snow cover with a pattern-based cloud gap filling approach. It is currently unclear whether seasonal snow depletion patterns are altered during drought, and whether cloud gap filling is negatively impacted. We analysed whether years of moderate severe drought alter patterns of snow depletion and reduce cloud gap filling reliability in the Boise River Basin, Idaho for the period of 2000–2024. We demonstrated moderate severe drought was uncorrelated with maximum snow extent, the onset of spring melt, and the rate of depletion. Patterns of snow depletion were similar at the watershed scale and robust to moderate severe drought (98.7% average correlation), and snowline representation is also highly similar (0.995, average R^2 over 68 models). Average cloud gap filling estimated similarity was 96.73% with a slight reduction during severe drought to 94.76%. Over one sixth of the world's population relies on water from snowmelt and real-time management of snowmelt runoff requires accurate snowline representation, which we accomplish with the dynamic seasonally recurrent pattern of snow depletion.

1 | Introduction

Seasonal snowpack is declining in the western United States in every state, month, and climate region (Mote et al. 2018). Snowmelt in the western United States contributes 50%–80% of the total freshwater (Stewart et al. 2004), and water management through accurate modelling is necessary for human demand and consumption in a changing climate (Mankin et al. 2015). Optical remote sensing of snow cover can provide daily data that can improve some snowmelt runoff modelling system accuracies (Steele et al. 2017; Di Marco et al. 2021; Parisouj et al. 2023). Cloud cover in optical remotely sensed

snow cover imagery obscures the ground surface, limiting our ability to detect snow. Cloud gap filling (CGF) methods infill cloud covered pixels (Li et al. 2019), but there is uncertainty in optically retrieved daily snow extent data that adversely affects snowmelt runoff modelling accuracy (Hussainzada et al. 2021). Non-optically based remote sensing is not affected by clouds, but these data lack the temporal resolution of optically based remote sensing (Margulis et al. 2016; Deschamps-Berger et al. 2022). Reliable cloud free daily estimates of snow cover in the western US are crucial, especially during drought (Dierauer et al. 2019), which is expected to have accelerated onset and increased intensity with increased

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climate variability (Mukherjee et al. 2018). Management and modelling of snowmelt runoff during drought are further stressed because average snow depletion timing is monotonically moving earlier at a rate of 0.44 days per year (Roessler and Dietz 2022). Snowmelt-fed streamflow is shifting earlier (Barnett et al. 2005; Mote 2006), which may reduce the resilience of the system to drought. Importantly, drought may reduce snow cover duration, shrinking the window for optimal observation of snow cover that may introduce errors in CGF methods.

Cloud gap filling is the process of infilling cloud covered pixels retrospectively or in real-time by spatial (Parajka and Blöschl 2008), temporal (Hall et al. 2019), multistep processes (Gafurov and Bárdossy 2009; Li et al. 2017), or mathematical relationships (López-Burgos et al. 2013). Uncertainties in snow location beneath cloud cover—especially the snow-to-snow-free interface referred to as the snowline hereafter—may vary seasonally and by cloud persistence. Snow location uncertainties may originate from wind-based redistribution of snow (Winstral and Marks 2002), extreme snowfall events (Lute and Abatzoglou 2014), or variations in climatic forcings (Clark et al. 2001; McCabe and Dettinger 2002). Uncertainty in snow location may propagate into CGF methods, negatively impacting the reliability of the resulting CGF dataset. The relationship between climate variability, drought, and snow is complex (Brown and Mote 2009; Viviroli et al. 2011), emphasising the need for reliable CGF snow cover methods and resulting datasets. Repeatable patterns of snow accumulation and depletion have been observed over multiple landscapes (Adams 1976; König and Sturm 1998; Luce and Tarboton 2004; Wang and Xie 2009; Sturm and Wagner 2010; Parr et al. 2020; Pflug and Lundquist 2020; Kiewiet et al. 2022). Snow depletion patterns have demonstrated accuracy greater than 84.9% for CGF (Woodruff and Qualls 2019); however, the variability introduced by drought may limit the effective implementation of pattern-based CGF.

CGF datasets provide a continuous time series of snow cover that may increase modelling accuracies for snowmelt runoff models (Steele et al. 2017; Di Marco et al. 2021; Parisouj et al. 2023). Snow cover accuracy directly impacts other critical variables like albedo, which is one of the most important factors for modelling snowmelt runoff (Blöschl 1991). Albedo affects the dominant driver of snowmelt: absorbed solar radiation and accurate estimation is crucial (Molotch et al. 2004; Dingman 2008; Bair et al. 2019; Ban et al. 2023). With cloud free data, snow properties including grain size, mixed pixels, and snow cover in forested areas can be extracted (Painter et al. 2009; Rittger et al. 2020). For CGF snow cover estimates, the snowline location is most important, especially for patchy or complex interfaces because it directly influences large contributors to snowmelt like sensible and latent heat fluxes (Harder et al. 2017; Schlögl et al. 2018).

Drought may influence maximum snow extent, rate of depletion, and the spatial progression of snow depletion that may introduce uncertainty in CGF methods. Generally, drought and snow research has focused on integrating snow dynamics like depth and snow water equivalent (SWE) into drought indices (Staudinger et al. 2014; Zhang et al. 2019), classifying snow

drought (Hatchett and McEvoy 2018; Dierauer et al. 2019; Huning and AghaKouchak 2020; Hatchett et al. 2022; Li and Wang 2022; Roberts-Pierel et al. 2024) and in some studies it is assumed that lower snow-covered area (SCA) leads to drought (Aghelpour et al. 2020; Pei et al. 2024). Maximum snow coverage is reduced by climatic controls like warming temperatures (Bormann et al. 2018), and relationships between SCA and drought vary regionally (Aghelpour et al. 2020; Safari et al. 2024). Seasonal snow cover is temporally dynamic with rapid reductions in coverage over short periods (Dye 2002), and the timing and duration of snow depletion vary year-to-year and over decades (Brown and Robinson 2011). Assessing the interaction of drought with these larger time scale shifts in snow cover is complicated. Quantifying the uncertainty drought introduces is pertinent because drought is occurring more often, for longer durations, and more intensely (Huning and AghaKouchak 2020) and the complex relationship between drought and snow cover is anticipated to become less predictable under future climate scenarios as snowpack declines (Livneh and Badger 2020).

Many aspects of snow depletion are altered by droughts and these alterations to timing and duration of snow cover may introduce uncertainty in mapped snow location for many CGF methods. For instance, regional snowline CGF approaches determine snow free, transition, and snow covered elevations to infill cloud covered pixels (Krajčí et al. 2014). If accumulation of snow was higher in one portion of the watershed in one year than in most other years, then the transition elevation will be larger in some portions of the watershed leading to an exceptionally large area of uncertainty. The regional snowline approach is common in many multistep CGF methods (Gafurov and Bárdossy 2009; Paudel and Andersen 2011; Da Ronco and De Michele 2014). Additionally, many multistep CGF methods employ a monthly probability of snow cover (Dong and Menzel 2016; Gafurov et al. 2016), which may not be reasonable in drought conditions where snow depletion timing may fluctuate out of the normal or average bounds. Many CGF approaches rely on consistent behaviour of snow depletion, rapid melting of a shallow snowpack and reduced SWE associated with drought (Dierauer et al. 2019). Rapid snowmelt provides fewer observation opportunities even if cloud cover is reduced. Therefore, during drought the importance of accurate coverage is paramount because regionally, snow coverage can disappear weeks earlier than normal years limiting the number of observation days. Any strong influence of drought on snow cover depletion may introduce significant reductions in CGF reliability for multiple commonly used CGF methods, and the uncertainty is poorly quantified.

Locational accuracy of the snowline in CGF datasets is paramount. The advection near the snowline can contribute a significant amount of energy to melt snow especially in patchy snow with longer snow to snow free interfaces (Mott et al. 2011; Harder et al. 2017; Schlögl et al. 2018). While watershed scale accuracy of CGF methods is widely discussed (Da Ronco and De Michele 2014; Krajčí et al. 2014; Dong and Menzel 2016; Gafurov et al. 2016; Li et al. 2019), explicit snowline locational accuracy is rarely quantified. Standard accuracy assessments include snow cover overall accuracy, overestimation, and underestimation (e.g., Deng et al. 2024). The generally large study regions, like high mountain Asia (Li et al. 2020; Hao et al. 2022), used for

CGF testing and validation may maintain high watershed scale accuracy while still producing localised errors in the snowline location. CGF approaches that use the explicit snowline to in-fill cloud covered pixels are preferred if the snowline follows a seasonal pattern and is unaffected by drought. The dynamic seasonally recurrent snow depletion pattern uses the explicit snowline to CGF (Woodruff and Qualls 2019); however, it is untested whether the snowline exhibits the same highly repeatable pattern for drought and non-drought conditions.

The ongoing threat of increased frequency, intensity, and duration of drought (Huning and AghaKouchak 2020) and intrinsic global importance of water from snowmelt requires reliable and rapidly available cloud free snow extent data. The goal of this research is to provide a cloud free, accurate, and dynamic estimate of daily snow cover with a pattern-based CGF approach. The critical problem with drought and reduced snowpack is shallower snowpack leads to greater variability in snow depth (Neumann and Marsh 1998; López-Moreno et al. 2011) and greater spatial variation in snow depth may drastically change the snow depletion spatial progression. Our hypothesis is that during drought the shallower snowpack will have a more complex interface between snow and snow free pixels negatively impacting CGF. We approach this hypothesis by first determining whether drought alters the spatial pattern of snow depletion used for CGF then assess impacts on the reliability of CGF. Our objectives are to (1) investigate snow depletion descriptive variables and their relationship to drought in the Boise River Basin; (2) determine if drought alters the CGF model similarity at the watershed and the snowline scale by comparing candidate models derived from drought years and drought free years; and (3) compare CGF reliability for drought years and drought free years.

2 | Data and Methods

2.1 | Study Site

The Boise River Basin is a large basin (6959.8 km²) situated to the Northeast of Boise Idaho, Figure 1. The basin is comprised of hydrologic unit codes 17050111, 17050112, and 17050113. Streamflow in the Boise River Basin is highly regulated by multiple dams (Lucky Peak, Arrowrock, and Anderson Ranch, Figure 1) and downstream diversions that serve as an important water resource for agriculture, hydroelectric power generation, recreation, municipalities, and in-stream flow requirements to support fish. Elevation ranges from 888 to 3202 m according to the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) Global Digital Elevation Model Version 3 with a spatial resolution of 1 arc sec or about 30 m (NASA 2019). The Boise River Basin received complete snow coverage three out of every 4 years during our study period: 2000–2024. Lower elevation snow in the study site melted quickly in late January or February, behaving like ephemeral snow; therefore, maximum snow extent was achieved by January 1. Mean annual precipitation for 2000–2016 was 90.0 cm, mean basin wide SWE was 59.3 cm, and mean annual temperature was 4.1°C (US Department of Agriculture 2025). Drought in the Boise River Basin has direct consequences to agricultural water users, making it a relevant test site.

2.2 | Data Sources

Two data sources were used in this study: the US Drought Monitor (USDM) and the Moderate Resolution Imaging

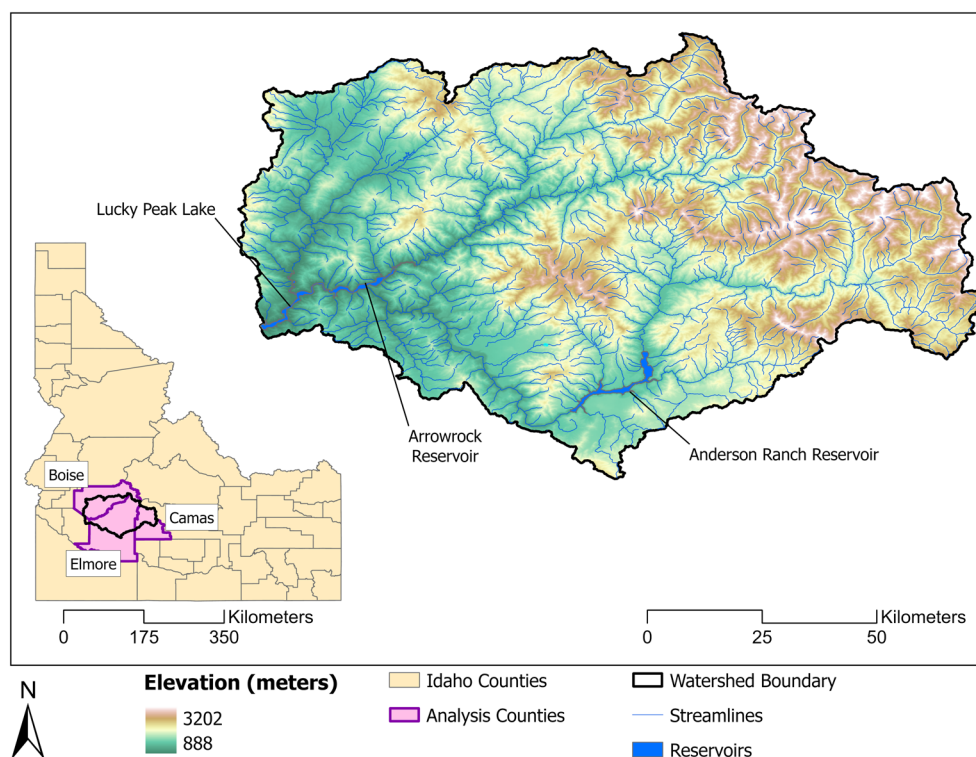


FIGURE 1 | The Boise River Basin boundary (black outline) study site is mapped with the digital elevation model derived from the ASTER 1-arc sec version 3 product, geographic location in Idaho (peach), counties (pink), streamlines (blue), and reservoirs (blue).

Spectroradiometer (MODIS). We used the USDM to classify drought and MODIS daily snow cover product to develop candidate models. We downloaded the USDM historic data for the period of 2000–2024 for three counties that covered the Boise River Basin: Boise, Elmore, and Camas counties, shown in Figure 1. The USDM classifies drought based on a mixture of data sources including physical indicators, field-based data, and state professional and local stakeholder knowledge. At the national level, the USDM calculates six physical indicators: Palmer Drought Severity Index (Palmer 1965), CPC Soil Moisture Model Percentiles (Huang et al. 1996), U.S. Geological Survey daily streamflow percentiles, Percent Normal Precipitation (Werick et al. 1994), Standard Precipitation Indices (McKee et al. 1993), and remotely sensed satellite vegetation health index (Kogan 1995). In Idaho, precipitation data is measured at the U.S.D.A. Natural Resource Conservation Service Snow Telemetry (SNOTEL) stations. Precipitation metrics include observations of SWE and percent normal precipitation.

The Idaho Drought Committee (IDC) is comprised of professionals throughout the state who also gather information from local end-user stakeholders. The IDC meets regularly to assess drought conditions and makes recommendations to the USDM regarding the spatial distribution and severity of drought across Idaho. The IDC makes its recommendation based on combinations of SNOTEL station SWE observations and statistics, NOAA National Water Prediction Service gridded maps of percent departure from normal precipitation over various preceding time periods (water.noaa.gov), U.S.G.S. streamflow, and/or other relevant variables. The USDM prohibits use of forecasts in drought assessment, relying on current and antecedent conditions instead. Additionally, the drought classification at any given location cannot be changed by more than one drought category in any given week. Both the national USDM and the IDC place more emphasis on one factor or another depending on the identification of the relative impact of each factor, which varies from year to year. This is necessary owing to the complex nature of drought and means there is a certain amount of art required in assessing drought rather than purely formulaic methods.

As a result of the USDM and IDC drought classification methods, it is springtime or even earlier conditions that strongly influence the evolution of drought conditions. USDM drought classifications may start in the spring but become more robust through the summer. Consequently, June–September drought categories robustly reflect conditions throughout the spring snowmelt period. We analysed June–September to assess drought conditions for each year. We reference drought to a particular year in order to associate it with a particular snowmelt year even though late spring through summer is the most relevant time period with respect to the snowmelt pattern and drought.

We used snow cover data collected by the MODIS sensor on board the Terra satellite (Hall and Riggs 2015). The MODIS sensor on board the Aqua satellite had a critical band six failure, which is used in the calculation of the Normalised Difference Snow Index (NDSI) for snow detection. Even though a solution was developed to address the Aqua band six failure (Shen et al. 2014), the Terra satellite MODIS sensor produces fewer snow cover omission errors (snow covered areas incorrectly identified as snow

free) than the Aqua satellite sensor (Hall et al. 2019). The Aqua and Terra satellites could be used collectively to reduce cloud cover (Parajka and Blöschl 2008); however, a robust seasonally dynamic snow depletion pattern can be extracted with only the Terra satellite (Woodruff and Qualls 2019). We split our analysis into a training set for the period of 2000–2016 (70%) and testing set 2017–2024 (30%). The training set included all available version six MOD10A1, MODIS daily NDSI snow cover, and the data was downloaded for the period of 2000–2016. The testing set included daily MOD10A1F version 6.1 data for 2017–2024. From the MOD10A1F dataset we used two layers: the CGF data and the NDSI snow cover data which contains cloud cover (Hall et al. 2019). The MOD10A1F NDSI snow cover data is equivalent to the MOD10A1 NDSI snow cover data. The NDSI snow cover is a product derived from raw NDSI. NDSI snow cover is analysis-ready and detectable snow is mapped over the range of 10–100 (Riggs et al. 2015). At the pixel level, we converted NDSI snow cover to binary classification: snow (values greater than or equal to 10) and no snow (values less than 10).

2.3 | Methods

2.3.1 | Classifying Drought

The USDM is produced weekly and provides five levels of drought D0 through D4 where D0 is abnormally dry and D4 is exceptional drought (Svoboda et al. 2002). We relabelled the classes to No Drought (0), Abnormally Dry (1), Low (2), Moderate (3), Moderate Severe (4), and Severe (5) for quantitative analysis and readability. For the years included in the study period, we needed to choose a means of identifying what should be considered drought. One of the significant issues we faced in classifying drought was the scarcity of moderate to severe drought. Nevertheless, significant drought occurred, and June through September provided the most robust classification of drought despite the complexity and interaction of multiple indicators. Our method to identify drought was as follows: The USDM is calculated weekly at the county level. Drought is assigned as a percentage of area, for example, 25% of the county was abnormally dry and 75% of the county was low severity drought. We generalised the county level data to a single class with preference to the highest level of drought recorded in a county. For example, if any portion of a county contained the class severe, the weekly value was recorded as severe. Weekly drought conditions were then aggregated to monthly values by assigning the monthly drought index as the mode of weekly drought. In the case of a tie, for example, 2 weeks at abnormally dry and 2 weeks at low severity, the higher severity was chosen. The simplification method was essential for prioritising the outcome of drought at the county level where some portions of the counties were outside of the headwater watershed, but surface water users downstream of the headwaters directly feel the impacts of drought. We calculated the above drought index for Boise, Elmore, and Camas counties, which fully cover the area of the Boise River Basin. Additionally, we defined a drought year as a year where moderate drought or greater is recorded in all three counties and at least 1 month of moderate severe drought was recorded. In the end, our methodology identified four out of the first 17 years and five out of the entire 25-year study period as the most significant drought years, or about 20% of the years.

2.3.2 | Qualitative Assessment of Snow Depletion and Drought Relationship

We ranked the drought years by averaging the monthly drought index values (0 for no drought to 5 for severe drought) across the three counties and summing the values from the 4 months, June through September, to get a summertime aggregated drought score. For example, if Boise County was abnormally dry (1), Elmore County was low severity (2), and Camas County was no drought (0) in June 2000 we would get a June average of 1 plus 2 plus 0 divided by 3 or 1. We compared the summer drought score with the watershed scale maximum snow extent, day of year (DOY) of 90% SCA, the rate of melt (SCA%/day) for the melting period of 90%–50% SCA, and the rate of melt for the melting period of 50%–0% SCA. We calculated correlations between the drought score and snow depletion metrics and SWE from the three highest elevation SNOTEL sites (Dollarhide Summit, Trinity Mountain, and Vienna Mine) to assess the degree of relationship between snow extent, melt rate, SWE, and drought. If a relationship between drought and snow cover metrics is observed, we would be motivated to derive separate drought specific models of the dynamic seasonally recurrent snow depletion pattern. If no relationship is observed between aggregated descriptive values of snow depletion and drought, the potential for spatial errors during drought still exists, and these errors must be quantified. We limit this analysis to 2000–2016 because we are determining whether drought impacts model development which occurred over the period of 2000–2016.

2.3.3 | Dynamic Seasonally Recurrent Snow Depletion Pattern Modelling

The dynamic seasonally recurrent snow depletion pattern can be derived from the collection of daily MODIS snow cover data spanning multiple years (Woodruff and Qualls 2019). For each year of record, the first observed date that each pixel transitioned from snow covered to snow free was extracted and saved into a seasonal composite. The seasonal composite, referred to as the First Day of Land (FDL), described the DOY of the first observed snow free day for each pixel starting on January 1. We calculated an FDL for each year 2000–2016. FDL composites extraction was used as a data preprocessing step to collect the relevant melt out information for each pixel into a seasonal flat raster format. An FDL composite is a raster map where each pixel or cell is attributed with its FDL for that calendar snow year (1 January to 31 December).

Ephemeral snowfall often occurred and re-covered previously snow-free areas. FDL composites only contained the first day each pixel was observed to be free of snow, ignoring subsequent ephemeral snow cover. Our work seeks to quantify the dynamic seasonally recurrent pattern of melt that is primarily driven by fixed watershed characteristics such as elevation, slope, aspect, radiation exposure, and so forth. We expect these characteristics to manifest themselves in the pattern revealed by the first time that each pixel melts off. Subsequent ephemeral snowfall occurs as spatial and temporal stochastic processes that exhibit random behaviour which differ from 1 year to another and have limited annually recurring relationships

with each other and with fixed watershed characteristics. In the work of Woodruff and Qualls (2019), the occurrence of ephemeral snowfall would cause a slight pause in the progression of the snowmelt pattern, but the new snow rapidly melted, and the snowline quickly returned to the previously observed pattern. Our model largely eliminates the “pause” effect of stochastic ephemeral snowfall. Consequently, it is the FDL values that are most relevant for extraction of the dynamic seasonally recurrent snow depletion pattern, and this justifies ignoring ephemeral snowfall.

The FDL composites were then transferred from their raster format to a vector of M values which corresponds to the M pixels in an FDL raster. Then the vector was saved to a tabular matrix with N years, one for each FDL raster with final dimensions of M by N . Additionally, we maintained the pixel centroid x, y coordinates in two columns of the matrix. The potential uncertainty cloud cover introduced into the derivation of the dynamic seasonally recurrent snow depletion pattern is negligible due to the processing steps taken in the sections below (Woodruff and Qualls 2019). Woodruff and Qualls (2019) demonstrated the FDL and counterpart composite—the last day snow is seen (LDS)—produced models of the dynamic seasonally recurrent snow depletion pattern that were 99.8% correlated to one another. The difference between the FDL and LDS was the number of cloud covered days between observations (Woodruff and Qualls 2019). Consequently, the methods used by Woodruff and Qualls (2019) eliminated the stochastic influence of cloud cover from the resulting snowmelt pattern.

We used principal component analysis (PCA) to derive the dynamic seasonally recurrent snow depletion pattern as demonstrated by Woodruff and Qualls (2019). PCA was applied to the multi-year tabular matrix of FDL composites (excluding x, y coordinate information). The output of PCA is eigen values and vectors. The eigen vectors (E) were multiplied by the input data (FDL tabular matrix) to transform the data into uncorrelated orthogonal Principal Components (PC's). Equation (1) describes the rotation of the FDL tabular matrix (M by N) multiplied by the eigen vector (E) to produce a principal component (PC1 is shown). In the case of snow depletion, the first principal component captures the dynamic seasonally recurrent snow depletion pattern (Woodruff and Qualls 2019). All other principal components were discarded. We rasterized the first principal component to the MODIS sinusoidal grid with the saved centroid data (FDL tabular matrix). The result was a PCA model of the dynamic seasonally recurrent snow depletion pattern in a raster format equal in spatial resolution to the FDL composites and MODIS data (500 m). The number of FDL composites (N) in Equation (1) was previously tested at a value of 17 (Woodruff and Qualls 2019); however, in this work we systematically reduce N as detailed below.

$$\begin{bmatrix} DOY_{1,1} & \cdots & DOY_{1,N} \\ \vdots & \ddots & \vdots \\ DOY_{M,1} & \cdots & DOY_{M,N} \end{bmatrix} \times \begin{bmatrix} E_1 \\ \vdots \\ E_N \end{bmatrix} = \begin{bmatrix} PC1_1 x_1 y_1 \\ \vdots \\ PC1_M x_M y_M \end{bmatrix} \quad (1)$$

The PCA mimics a T-mode PCA often used in atmospheric applications (Raziei and Richman 2025). We preserved the seasonal pattern of snow depletion for each year (columns, Equation (1)) by scaling them with PCA analysis. In Equation (1), FDL dates go in

as input and the melt order comes out as PC1. Values closer to zero melt before values further from zero. Seasonal patterns of snow depletion are largely deterministic and the ordered output from Equation (1) describes pixel melt order, which was the seasonal depletion pattern.

2.3.4 | Watershed Scale Similarity Sensitivity Analysis

Drought may impact the rate and extent of snow depletion potentially introducing uncertainty into the FDL composites. Large deviations in the FDL composites from “normal behaviour” may propagate into the dynamic seasonally recurrent snow depletion pattern affecting the order of pixel melt described by the PCA. We tested the spatial similarity of the dynamic seasonally recurrent snow depletion pattern by comparing deviations of multiple candidate models compared to PCA_{17} . PCA_{17} was the dynamic seasonally recurrent snow depletion pattern derived by combining 17 FDL composites (2000–2016). From Equation (1), N equals 17 which produced the dynamic seasonally recurrent snow depletion pattern for the Upper Snake River Basin in Woodruff and Qualls (2019).

We calculated deviation from PCA_{17} as correlations of a candidate model (PC1, Equation (1)) to PCA_{17} (PC1). We performed a sensitivity analysis where (1) we calculated candidate PCA models derived by varying the number of FDL composites used (Q , where Q ranged from 3 to 16 in increments of 1), (2) for every value of Q we calculated the correlation of the candidate PCA models to PCA_{17} , and (3) we analyzed the value of the correlation and the frequency of FDL composites associated with drought in the candidate model to determine if drought led to lower candidate model similarity to PCA_{17} .

For each value of Q , we randomly sampled from the list of possible combinations of FDL composites, or simply the list of years (2000–2016). For example, for a $Q=5$, in a set of 17 unique numbers (one for each year), there exists 6188 possible combinations. In Equation (2), combinations (C) were calculated as a factorial of the maximum number of FDL composites, 17. A practical minimum Q of 3 was selected for principal component analysis.

$$C = \frac{17!}{(17 - Q)! * Q!} \quad (2)$$

By Equation (2), the number of combinations changed for each value of Q . We first calculated and stored all possible combinations for a given Q . We then randomly sampled from the list of possible combinations where the sample size (n , Table 1) was calculated as 10% of the total number of combinations or a maximum of 100 to maintain balance between different values of Q . In the case of $Q=16$, we used all 17 possible combinations, Table 1. For each value of Q , the sample size (n) indicated the number of combination sets. For every combination set, we calculated a unique candidate PCA model.

As shown in Table 1, a total of 1167 candidate PCA models were generated (summation of column 3, sample size [n]). For each value of Q , we calculated the minimum, average, and maximum correlation of all candidate PCA models to PCA_{17} . Systematic reductions in model similarity due to drought were assessed

TABLE 1 | Possible combinations and sample size are shown for each value of Q , which determined the number of FDL seasonal composites used to develop a PCA candidate model.

Number of years used in combination (Q)	Total number of possible combinations	Sample size (n)
16	17	17
15	136	14
14	680	68
13	2380	100
12	6188	100
11	12 376	100
10	19 448	100
9	24 310	100
8	24 310	100
7	19 448	100
6	12 376	100
5	6188	100
4	2380	100
3	680	68

by plotting the correlations for all candidate models ($Q=3$) and colour coding the plotted data by the frequency of drought associated FDL composites that were used to derive the candidate model. A drought associated FDL composite was an FDL derived during a year where a drought year was classified by Section 2.3.1. We visually assessed whether correlations were systematically lower when drought FDL composites were used to develop the candidate PCA model.

2.3.5 | Pixel Based Interannual Snowline Similarity

Although watershed scale similarity is a good indicator of model performance, the actual locational accuracy of the snow-to-snow-free interface (snowline) is paramount for cloud removal (Woodruff and Qualls 2019). We assessed the interannual snowline similarity by ranking the pixels in each candidate model from first to melt to last to melt. A similar ranking between corresponding pixels in different candidate models meant the snowline was stable because the order of melt was maintained. We calculated the r-squared values of the ranked pixel melt order for each candidate model (developed for $Q=3$) compared with PCA_{17} . We calculated the 95% confidence intervals of the resulting r-squared values. If drought year candidate models recorded large spatial shifts in the snowline at any time step during the melting season, then the ranked assessment would show a reduction in the r-squared through either distinct shifts or greater variance. For example, let us consider two pixels on opposing geographical sides (east and west) of a watershed, where pixel A melted 100th and pixel B melted 101st. If drought altered the snowline location by accelerating snow depletion near pixel A, then many pixels would melt ahead of pixel B, and its observed melt rank would be shifted (e.g.,

200th). When plotted in the ranked assessment, pixel B would be (101, 200), which is quite different than (101, 101). Additionally, all other pixels that should have melted near the same ranked order as pixel B would have a larger melt rank. In a scatter plot comparison, the variability in the snowline may show up as bias about a one-to-one line. The ranked assessment directly quantifies the interannual snowline similarity throughout the entire melting period. Even if drought impacts were not apparent at the watershed scale, the ranked analysis could reveal uncertainties introduced by drought.

2.3.6 | Cloud Gap Filling

In the previous sections we determined whether drought directly impacts the derived PCA model. In this section we assess CGF. We calculated the similarity of snow cover estimate of CGF with the PCA model versus the MOD10A1F CGF dataset for images from 2017 to 2024. Our analysis focused on images during the melting period (DOY 1: 250). We filtered images (MOD10A1F and NDSI snow cover images) to only include imagery with watershed scale SCA between 90% and 10%. Finally, we filtered the images to only those with less than 75% cloud coverage because the CGF algorithm developed by Woodruff and Qualls (2019) relies on visible pixels in a cloud covered MODIS image.

The PCA model can be fit to an image by optimising a threshold value, τ . The PCA model ranks pixels from first to melt to last to melt. As τ moves further from zero, modelled SCA moves from

100% coverage to 0% coverage. For CGF, we converted the continuous PCA values into snow free (below τ) or snow covered (above τ). For a MODIS image, we select τ by optimising the visible pixel error (VPE), given in Equation (3). We first select a value of τ , convert the PCA model to a snow and snow free image and compare with the MODIS image which yields three values: the number of incorrectly modelled snow free pixels (I_B), the number of incorrectly modelled snow-covered pixels (I_A), and the total number of visible pixels (T_P), where incorrect is determined by comparing model produced pixel values with MODIS observed values. For the MOD10A1F CGF data, T_P is all of the pixels contained in the watershed boundary. For the NDSI snow cover data, T_P can be as low as 0 for completely cloud covered MODIS images. We slowly increased τ until the VPE is optimised to the minimum value. We then calculated the percentage cloud free pixels correctly mapped based on the visible cloud free pixels.

$$\text{VPE} = \sqrt{\frac{I_B^2 + I_A^2}{T_P^2}} \quad (3)$$

3 | Results

3.1 | Classifying Drought

The US Drought Monitor data is summarised by month for Boise (blue), Elmore (black), and Camas (red) Counties for 2000–2024 in Figure 2. The cumulative drought is shown in

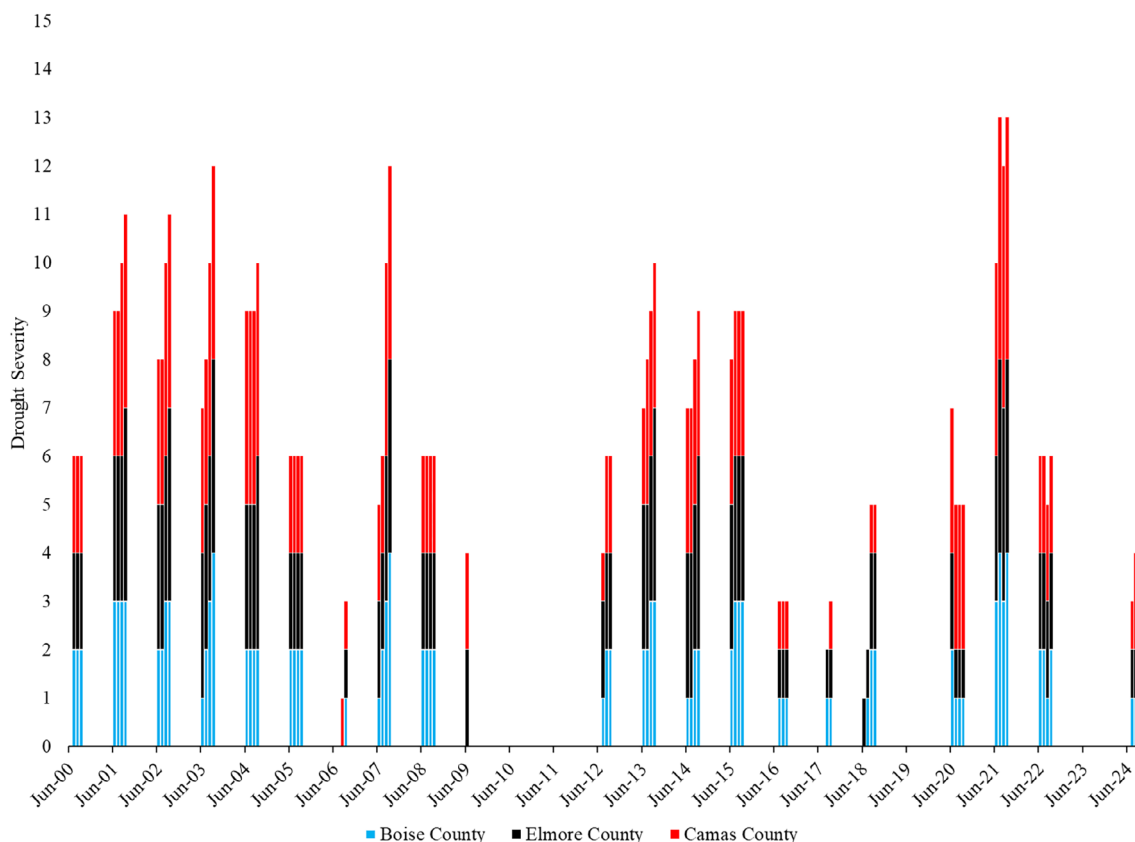


FIGURE 2 | Drought classes from 0 (no drought) to 5 (severe drought) are shown as a stacked bar for Boise (blue), Elmore (black), and Camas (red) counties. The cumulative drought class is given as the total height of each bar and the contribution by each county is given by the colour of bar. For each year, a bar for each month is plotted, June–September.

Figure 2. The drought value for each county—no drought (0) to severe drought (5)—for each month is plotted as a stacked bar. A maximum value of 15 indicates there was severe drought recorded in all three counties. The majority of the snowpack accumulates in the higher elevations of Boise and Camas counties. Camas County recorded the highest incidence of moderate severe drought. Severe drought was only recorded in 2021 and only in Camas County. Moderate severe drought was more common, and we can further classify these results and define them as drought years: 2001, 2002, 2003, 2013, and 2021 because these years contained moderate or greater drought in all three counties and at least 1 month of moderate severe drought. There were 6 years with little to no drought in all 4 months: 2006, 2009, 2010, 2011, 2019, and 2023. All remaining years fell in the middle somewhere between moderate severe drought and no drought.

3.2 | Qualitative Assessment of Snow Depletion and Drought Relationship

The drought monitor results were generalised into a drought score (Table 2), representing the average monthly summer-time drought severity per county, calculated as the cumulative annual (June to September) drought severity from Figure 2, divided by three counties. The drought score was compared

with maximum snow extent, DOY of 90% SCA, melt rate (SCA%/day) for 90%–50% SCA, the melt rate (SCA%/day) for 50%–0% SCA, and average maximum SWE across the three highest elevation SNOTEL sites located within the watershed. The descriptive snow statistics were chosen to represent important stages of snow disappearance but likely do not represent the entire snow regime. Two rows of Table 2 are shown in bold text: 2001, which had the highest drought score, and 2011, which was one of the years with no drought recorded. Both 2001 and 2011 had the same maximum extent, and similar late melt rates (50%–0% SCA); however, the DOY of 90% SCA was far earlier in 2001 (69) than 2011 (90), and the early rate of melt (90%–50% SCA) was much higher in 2001: 2.5% SCA/day versus 0.98% in 2011. The comparison of these 2 years appears to indicate that the early first half of the SCA melt rate and DOY of 90% SCA distinguish between high versus low drought scores, but this distinction is not generally observed. For example, 2002 had a drought score of 12.3 and had the same DOY of 90% SCA as 2011 with a drought score of 0. Correlations between drought scores and all variables in Table 2 were below 0.5. The low correlation of maximum SWE with the drought score is likely due to the composite and variable nature of drought classification from the USDM which integrates streamflow as well through a highly regulated watershed. Even though there is no readily apparent relationship between drought and snow depletion, the temporal aggregate

TABLE 2 | Yearly drought score derived by aggregating monthly drought for Boise, Elmore, and Camas counties is given with snow depletion metrics including seasonal maximum snow extent, day of year (DOY) of 90% snow covered area (SCA), melt rate in SCA % per day for 90%–50% SCA, melt rate in SCA % per day for 50%–0% SCA, and average maximum snow water equivalent (SWE) of the three highest elevation SNOTEL sites in centimetres. The two bolded rows represent examples of the highest and lowest drought scores.

Year	Drought score	Max extent	90% SCA day of year	Melt rate (90%–50%) SCA%/day	Melt rate (50%–0%) SCA%/day	Average maximum SWE (cm)
2000	6.0	100.00	73	1.14	0.64	73.49
2001	13.0	100.00	69	2.50	0.51	42.93
2002	12.3	100.00	90	1.74	0.68	77.98
2003	12.3	100.00	37	0.63	0.54	89.32
2004	12.3	100.00	77	2.11	0.54	68.66
2005	8.0	100.00	55	0.89	0.56	55.29
2006	1.3	99.98	100	2.22	0.66	119.21
2007	11.0	99.90	71	3.33	0.52	54.86
2008	8.0	100.00	103	1.54	0.72	85.26
2009	1.3	100.00	87	1.67	0.60	74.25
2010	0.0	100.00	82	1.21	0.65	61.64
2011	0.0	100.00	90	0.98	0.61	100.33
2012	5.3	99.95	82	1.38	0.56	84.58
2013	11.3	100.00	63	1.48	0.55	62.57
2014	10.3	99.63	59	0.93	0.53	85.94
2015	10.7	100.00	39	1.03	0.52	60.45
2016	3.0	100.00	62	1.05	0.57	87.88

statistics presented in Table 2 may overlook important spatial differences in snow depletion associated with drought.

Spatial differences in snow depletion at different points throughout the snowmelt season can be visualised by comparing the FDL composites for a drought year, 2003 (Figure 3a), and a non-drought year, 2011 (Figure 3c). The darker blue colours melt first, and the lighter blue colours melt last. For clarity, the FDL is a composite of the DOY each pixel melted and describes the seasonal history of each pixel's DOY of snow depletion for a given year. By selecting the same colour bar for both Figure 3a,c we are pseudo-ranking the pixels from first to melt to last to melt rather than comparing the actual day of melt, which would be problematic because in 2003 the DOY of 90% SCA was 37 and in 2011 it was 90. The gradation of blue over the entire watershed boundary appears quite similar for both FDL seasonal composites, especially for lighter blues (Figure 3). A more explicit method of comparing the FDL composites and the seasonal progression of snow depletion is to compare the snow coverage for a specific SCA. We mapped 40% SCA because it is near the middle of the snowmelt period and highlights differences between FDL composites for

2003 (Figure 3b) and 2011 (Figure 3d). In Figure 3b,d, solid light blue is snow covered, and brown is snow free. At first glance the snow coverage is exceptionally similar; however, closer inspection of the 2003 snowline (Figure 3b, indicated by the black arrow) demonstrates a more complex and patchier snowline when compared to the continuous snowline for 2011 (Figure 3d, indicated by the black arrow). The distribution of snow is also slightly different, specifically in the eastern end in 2003 versus 2011 (Figure 3b,d). There is spatial dissimilarity between the FDL composite for drought and drought free years. These subtle differences may propagate into the PCA analysis by altering the order of snow depletion in a candidate model. FDL correlations are given in Figure S1. All correlations were greater than 0.83, and the lowest correlation was recorded for 2003 versus 2004.

3.3 | Watershed Scale Similarity Sensitivity Analysis

Objective two of this research is to determine if drought alters the similarity of CGF models at the watershed and the snowline

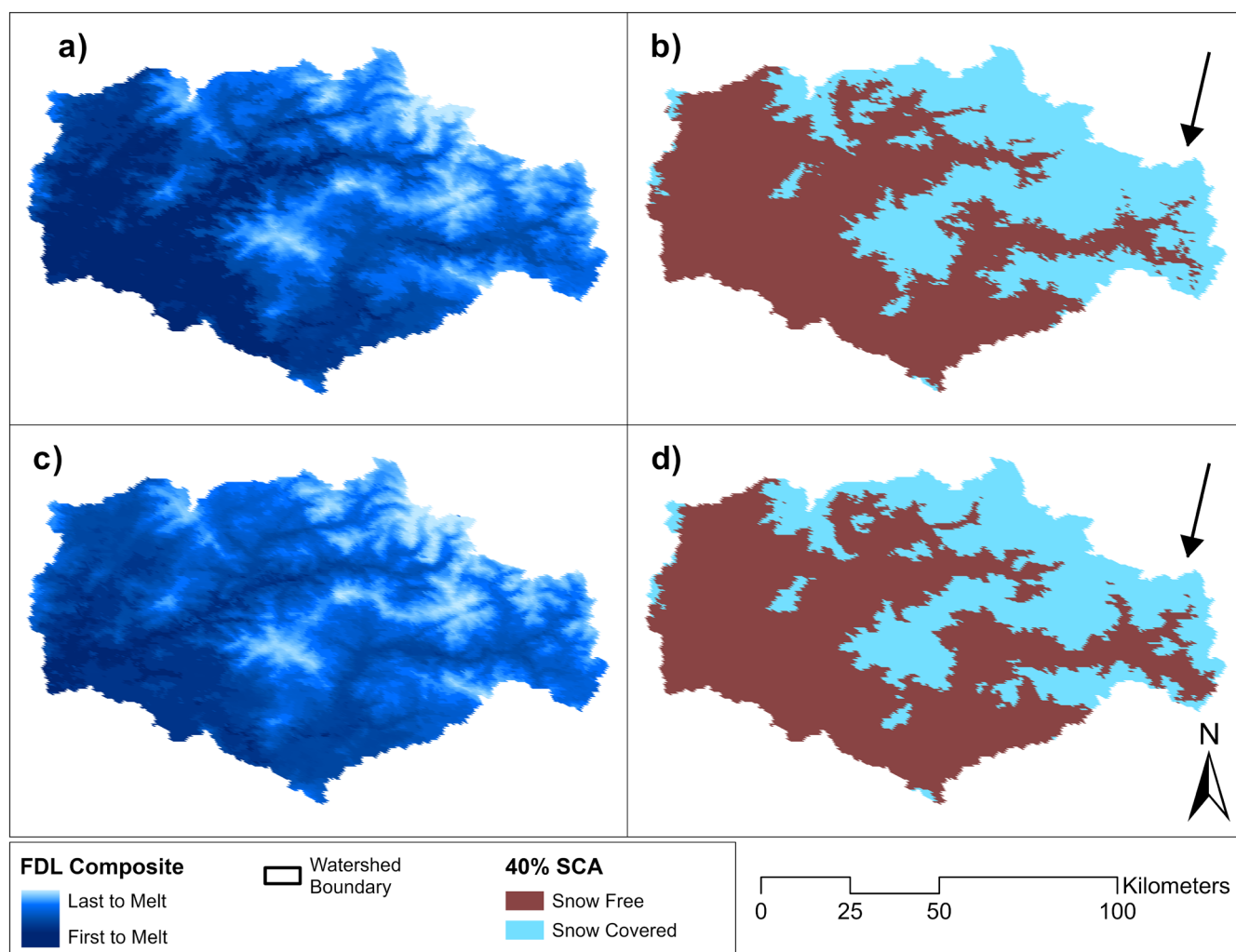


FIGURE 3 | The FDL seasonal composites for 2003 (a) and 2011 (c) are mapped as a gradient of blues where darker blues melted earlier than lighter blues. The FDL seasonal composites describe the full seasonal snow depletion progression for the watershed. Snow coverage (light blue) for 40% snow covered area for 2003 (b) and 2011 (d) are also provided as snapshots of snow coverage for both years. Note the patchier snow (blue) to snow free (brown) interface for 2003 compared to 2011.

scale by comparing candidate models derived from drought years and drought-free years. We determined the inclusion of FDL composites associated with drought did not reduce watershed scale similarity even when we reduced the number of FDL composites used to derive candidate models. For PCA_{17} , which contains the maximum amount of data, the proportion of total variance described by Principal Component 1 was 0.926. For Principal Component 2, the variance described drops to 0.018, meaning nearly all of the variance is described by Principal Component 1 (Table S1). These results agree with Woodruff and Qualls (2019), and we discarded the remaining principal components for the remainder of the analysis.

First, we conducted a sensitivity analysis to determine the impact of reducing Q on the correlation between candidate models to PCA_{17} . A total of 1168 candidate PCA models of the dynamic seasonally recurrent snow depletion pattern were created and evaluated. Figure 4 shows the resulting correlation statistics, minimum (red x), average (solid blue dot), and maximum (hollow black square) for each value of Q . With a Q of 3, average correlation to PCA_{17} was high: 0.987 and even the minimum correlation was high: 0.974. Maximum and average correlations steadily declined as Q decreased, but the minimum correlation was less consistent, suggesting the introduction of spurious error into some candidate models.

We then determined that the inclusion of FDL composites from drought years does not negatively or systematically reduce correlations by plotting the correlation of each candidate model for $Q = 3$, compared to PCA_{17} colour coded by the frequency of moderate severe drought associated FDL composites included in the candidate model, Figure 5. In Figure 5, red solid squares included two moderate severe drought FDL composites, blue hollow triangles included one moderate severe drought FDL composite, and solid black circles were derived from FDL composites that

were not associated with moderate severe drought. Three candidate models were developed with two moderate severe drought FDL composites, and their correlations exceed 0.985 (Figure 5, red squares). Of all the correlations plotted, only one candidate model fell below a correlation of 0.98. At the watershed scale, variability in candidate model similarity developed with as few as 3 FDL composites does not systematically decline. We observed the same response for all candidate models derived over the range of Q .

3.4 | Pixel Based Interannual Snowline Similarity

The second half of objective two is to determine if the snowline representation is altered by the inclusion of drought years in model development. We demonstrate the interannual snowline similarity is most variable for pixels ranked between 5000 and 15000 (maximum rank is 32381), and the variability between candidate models derived with a Q of 3 is greater than between any given candidate model and PCA_{17} . We quantified interannual snowline similarity by comparing the ranked pixel values or melt order of candidate PCA models ($Q = 3$) with PCA_{17} . A total of 68 candidate models were converted from their native PC1 values to ranked values and compared directly to the ranked PCA_{17} PC1 values. We calculated the r-squared of each of the 68 ranked candidate models versus PCA_{17} and found an average r-squared of 0.9954, with 95% confidence intervals of 0.9951 and 0.9957.

Based on the assessment of all 68 candidate models, it appears the snowline is extremely consistent; however, it is worth investigating how the snowline ranked pixel values compare between models derived with the same Q . We plot four examples to visualise the influence of FDL composites associated with drought as heat maps in Figure 6. In all four panels of

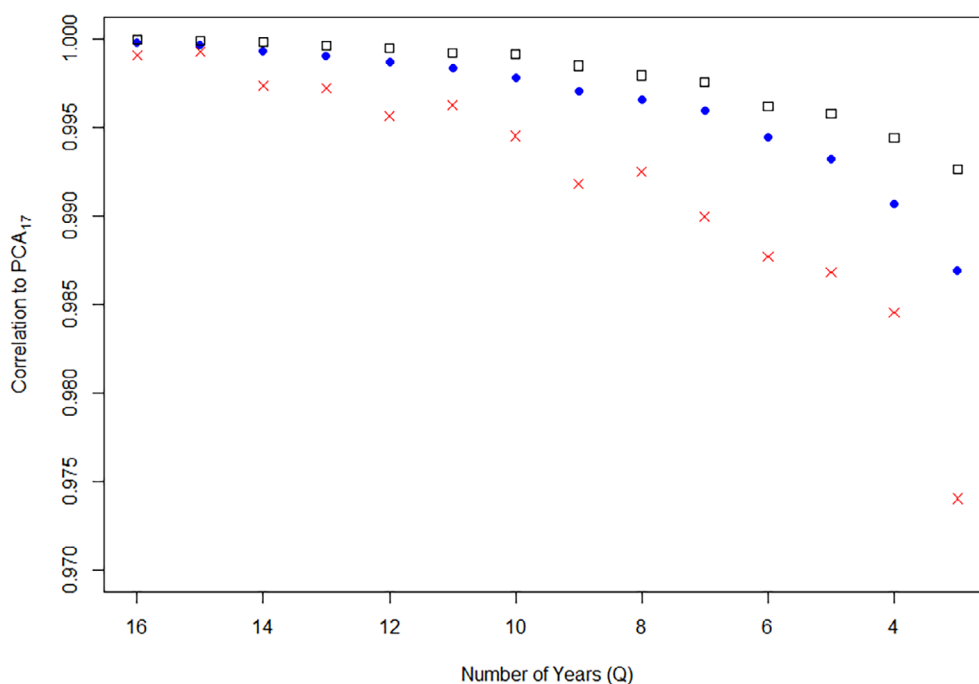


FIGURE 4 | Maximum (black hollow squares), average (solid blue dots), and minimum (red x) correlations of candidate PCA models compared to the PCA_{17} model are shown by the number of FDL composites (Q) used to develop candidate models.

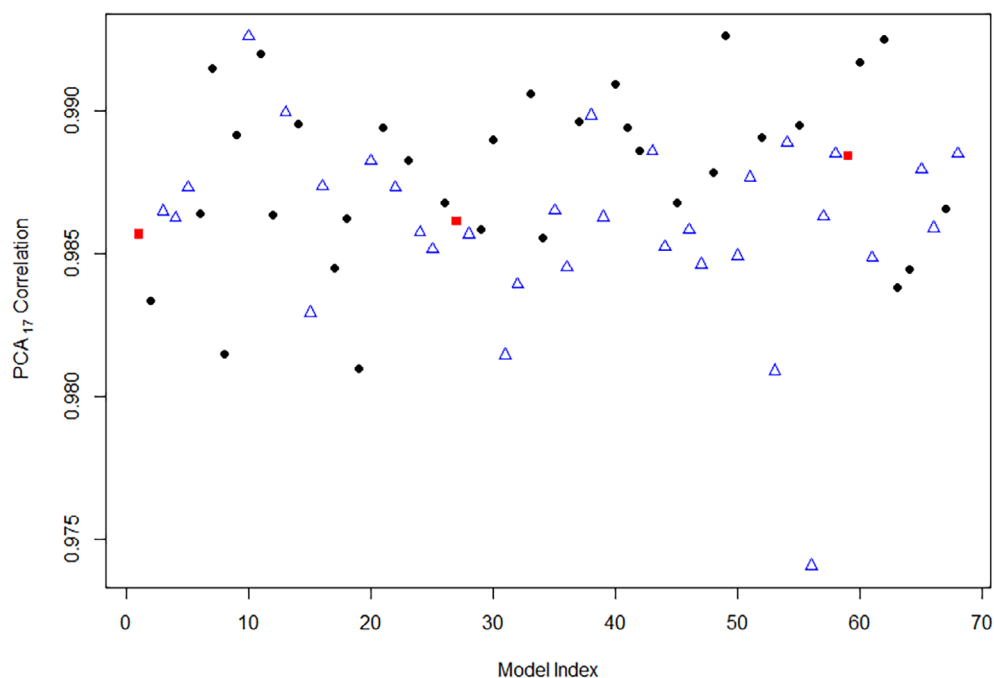


FIGURE 5 | Correlations are plotted for all candidate PCA models for $Q=3$ compared to PCA_{17} ($Q=17$). The x axis depicts candidate PCA model index which ranged from 1 to 68. Plotted correlations are categorised and colour coded as follows: Candidate PCA models developed with two drought associated FDL composites out of the three total FDL composites are plotted as solid red squares, one drought associated FDL composite as hollow blue triangles, and candidate models developed with FDL composites that were not associated with moderate severe drought are solid black dots.

Figure 6, the x axis is the ranked pixel values for the candidate model developed from FDL composites 2000, 2001, and 2003. Figure 6a plots on the y axis the ranked pixel values of a “drought” candidate model derived from FDL composites 2001, 2002, and 2009; Figure 6b an “average” candidate model, 2007, 2008, and 2014; Figure 6c a “wet” candidate model 2010, 2011, and 2016; and Figure 6d PCA_{17} . Figure 6a contains a comparison of models both developed with the inclusion of 2001 because our random sampling, Section 2.3.4, produced only three candidate models including at least two drought years, all three of which included some combination of common drought FDL composites. If the ranking or snowline was consistent for each candidate model comparison, the heat map would be most intense about the black 1:1 line, Figure 6. Figure 6a,b have the largest variability about the one-to-one line, which correspond to “drought” and “average” candidate models. Interestingly, the “wet” candidate model had less variability about the one-to-one line (6c), and the least amount of variability is observed in the comparison of a “drought” candidate model against PCA_{17} (Figure 6d). Season to season snowline ranked melt varies to some extent indicating the snowline is not consistently in the same location at each time step throughout the snowmelt period; however, there is no bias. At a low value of Q , the variability is greater between models of the same value of Q than between any given model and PCA_{17} .

3.5 | Cloud Gap Filling

Our final objective is to compare PCA_{17} -derived and MOD10A1F-derived CGF snow cover estimates for independent data years

2017–2024, colour coded by degree of drought severity for each year. Boxplots of snow cover pixel similarity are shown in Figure 7. Pixel similarity is the percentage of cloud free pixels that are classified the same between the PCA_{17} model and either MOD10A1F or the PCA_{17} model and the visible pixels in the MODIS image. The yearly similarity of the PCA_{17} fit to the MOD10A1F CGF product is given in Figure 7a, and yearly similarity for PCA_{17} CGF is given in Figure 7b. Boxes are coloured according to drought severity where blue is drought free, yellow is low or abnormally dry, orange is low to moderate, and red is severe. Snow cover similarities have a greater range shown by the larger box heights in Figure 7a compared to Figure 7b. PCA_{17} CGF snow cover similarity percentages are lowest during the severe drought year of 2021 with an average of 94.76%. The PCA_{17} CGF estimate similarity is larger than the PCA_{17} versus the MOD10A1F value of 94.36%. Overall snow cover similarity was 96.73% for the PCA_{17} CGF compared to 95.64% for the PCA_{17} versus the MOD10A1F CGF data. The lowest average similarity was 92.08% in 2022 for PCA_{17} versus the MOD10A1F CGF data.

4 | Discussion

Drought represents a regionally variable uncertainty for snow cover, which may introduce long term errors in CGF methods. Our objectives were to (1) investigate snow depletion descriptive variables and their relationship to drought in the Boise River Basin; (2) determine if drought alters the CGF model similarity at the watershed and the snowline scale by comparing candidate models derived from drought years and drought-free years; and (3) compare CGF reliability for drought years and drought-free years. We demonstrate the robust similarity

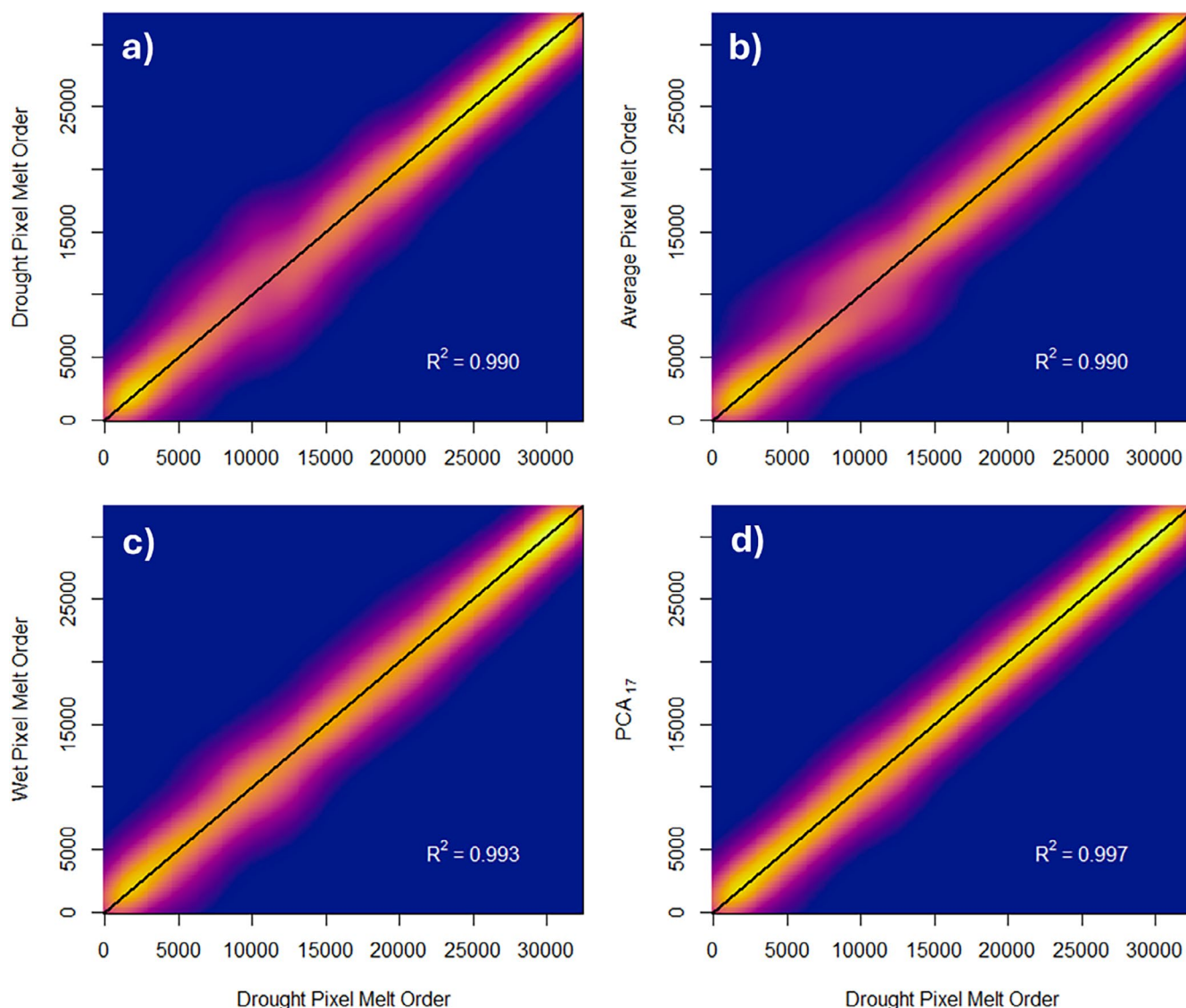


FIGURE 6 | Heat maps of ranked candidate model comparisons are shown with a black one-to-one line for reference. In all four panels along the x axis the ranked pixel values are plotted for a candidate model ($Q=3$) developed with two drought associated FDL composites 2000, 2001, and 2003. (a) Plots the ranked pixel values for a candidate PCA model developed with two drought associated FDL composites 2001, 2002, and 2009 ($r^2=0.990$). (b) Plots ranked pixel values for a candidate PCA model developed with average FDL composites 2007, 2008, and 2014 ($r^2=0.990$). (c) plots ranked pixel values for a candidate PCA model developed with wet year associated FDL composites 2010, 2011, and 2016 ($r^2=0.993$). (d) Plots the ranked pixel values of PCA_{17} ($r^2=0.997$).

of the PCA model to inclusion of drought. Average CGF snow cover similarities were greater than 94.75% for the PCA_{17} CGF and greater than 92.08% for the comparison of PCA_{17} and MOD10A1F CGF (Figure 7). Drought is rarely discussed in the context of CGF, even though spatial and temporal variance in snow cover and duration may be higher during drought. Increased variance in snow cover dates and coverage may adversely affect CGF with regional snowline and multi-year monthly snow cover probability methods that are commonly used (Gafurov and Bárdossy 2009; Paudel and Andersen 2011; Da Ronco and De Michele 2014; Krajčič et al. 2014; Dong and Menzel 2016; Gafurov et al. 2016). We demonstrate the dynamic seasonally recurrent snow depletion pattern, which relies on the order of melt, is flexible enough to accurately represent snow cover during severe drought.

Our study focused on one CGF method (Woodruff and Qualls 2019) and assessed the reliability of the model during drought conditions at the watershed and snowline scales. CGF with a pattern-based approach relies on the snowline (Equation (3)). Seasonal patterns of snow depletion are generally robust to drought, indicating a larger watershed scale truth about snowmelt that is highly reliable for CGF (Woodruff and Qualls 2019). An important advantage of CGF with the dynamic seasonally recurrent snow depletion pattern is the spatial progression of snow cover is flexible, even under the large variability in snow depletion timing. The dynamic pattern is a function of the stable drivers of snowmelt like elevation, aspect, slope, vegetation, and (given time of year) incoming solar radiation (Hock 2003; Dingman 2008). While at a large scale we note high interannual similarity, at the local snowline

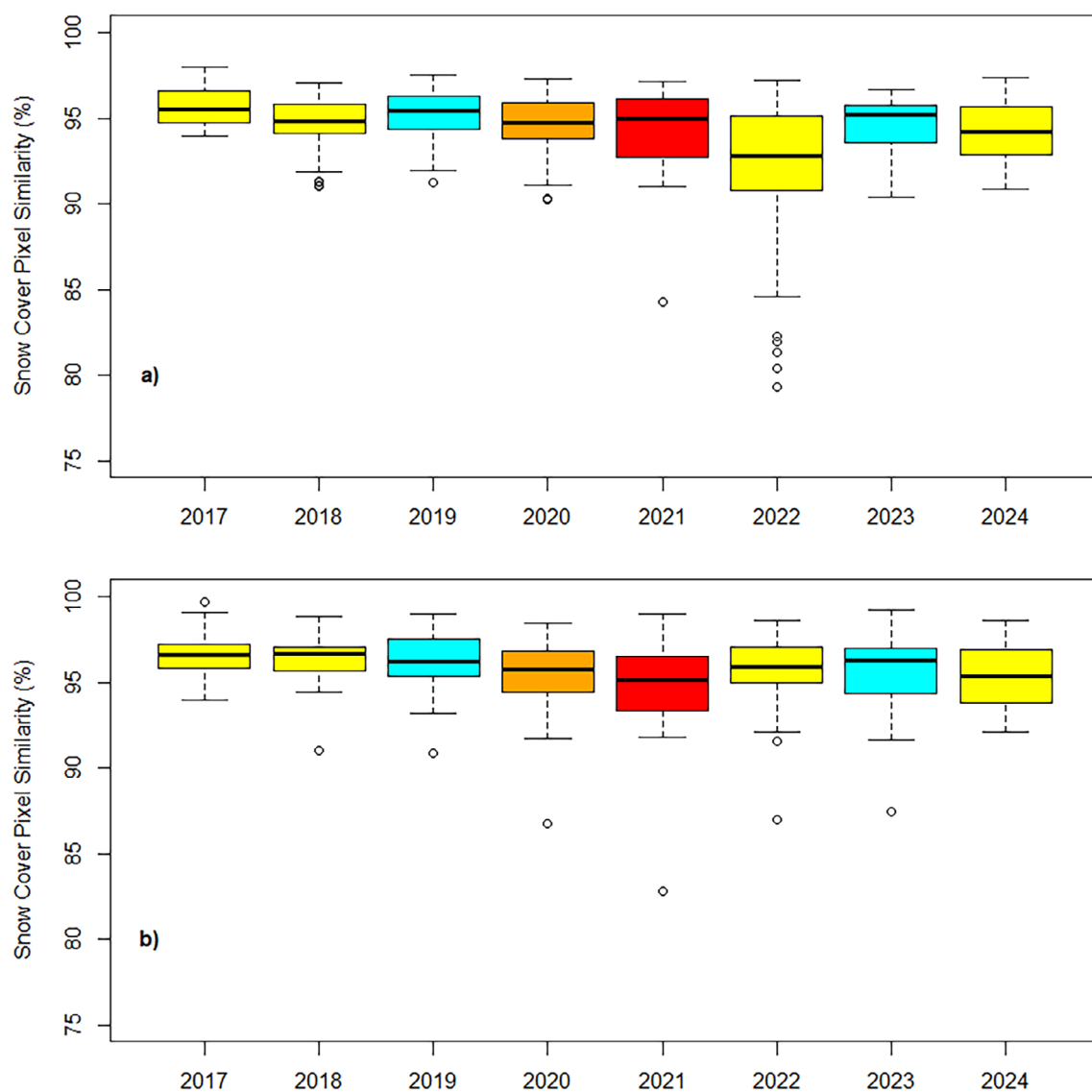


FIGURE 7 | Boxplots of snow cover pixel similarity are given by year for PCA₁₇ versus the MOD10A1F CGF product (a) and PCA₁₇ CGF (b). Boxes are colour coded according to drought severity where blue is no drought, yellow is abnormally dry to low, orange is low to moderate, and red is severe. Percentages were calculated during the melting period for images with watershed SCA of 90%–10% and with less than 75% cloud coverage.

level, some spatial variability is introduced by drought. We noted a small drop in CGF snow cover similarity during the 2021 drought which can be attributed to a more complex and patchier snowline. The local snowline scale differences may be a function of a shallow snowpack which increases spatial variability in snow depth (Neumann and Marsh 1998; López-Moreno et al. 2011). Although some have demonstrated shallow snow leads to increased snow depth variability, Prasad et al. (2001) showed that a large change in precipitation input dramatically shifted the snow depth distribution function but yielded a minimal shift in the relative spatial distribution of snow when SWE is normalised by the station precipitation. Other potential reasons could be poor snow and snow-free discrimination by the MODIS sensor, which occurs in forested areas (Rittger et al. 2020).

A critically important finding of the study is the high similarity of the dynamic seasonally recurrent patterns of snow depletion as Q was decreased. Candidate models derived with only three

FDL composites were highly correlated (0.987) to a model developed with 17 FDL composites. The high correlation may be attributed to overlapping input data; however, the interannual similarity in the rank-based models developed with a Q of 3 produced an R^2 of 0.9954 and many of the 68 candidate models did not contain overlapping data. The high correlation is important for CGF because the model validity can only be assessed over unseen data, for example, 2017–2024 in our study. Minimising the number of FDL composites needed to develop the PCA model (training data for model development) increases the number of years for unbiased CGF accuracy assessment (testing data). Importantly, the ability of the dynamic seasonally recurrent snow depletion pattern to predict the spatial progression of snow depletion has significant implications for snowmelt runoff modelling. Accurate representation of the snowline is critical for modelling advection (Schlögl et al. 2018), turbulent fluxes (including boundary conditions over snow) (Haugeneder et al. 2024), and albedo (Dingman 2008). As drought and dry periods are expected to be hotter, last longer, and reduce the

fraction of the overall precipitation expected to fall as snow (Barnett et al. 2008; Luce et al. 2016), water management and modelling with reliable snowline estimates has increasing value.

The rank-based assessment of pixel ordering by different candidate models in Section 3.4 demonstrates the interannual similarity between candidate models even under varied drought conditions. In the Boise River Basin, snow depletion from 100% to 0% SCA occurs over ~100 days. Thousands of pixels melt each day during periods of rapid melt. We noted variability in the rank candidate model pixels indicating variability in the order of melt and subsequently interannual snowline repeatability. The sheer number of pixels converting from snow covered to snow free during active melt each day reduces the overall importance of a single pixel's ranked melt order so long as the change in rank does not exceed the number of pixels that convert on a single day. If drought truly reduced CGF reliability or model performance, the model similarities during drought would have recorded larger errors than drought free years, which we did not observe during 2017–2024. We do notice larger discrepancies between the PCA_{17} model and MOD10A1F, which is likely due to the CGF method employed by MOD10A1F. The MOD10A1F product is backward-looking and brings the last cloud free observation forward, favouring over-estimation of snow cover during the melting period. The difference in CGF methods may be the source of the CGF similarity difference in 2017–2024.

4.1 | Limitations

In this study we used the daily MODIS snow cover extent (MOD10A1 and MOD10A1F), which produces a NDSI snow cover value that was converted to binary snow or snow free. For development of the dynamic seasonally recurrent snow depletion pattern, it is unclear if the steps presented here will seamlessly transfer to higher spatial resolution sensors because of the reduction in temporal resolution. While temporal resolution concerns are valid, cloud cover already obscures pixels for multiple days (Hall et al. 2019), and the model results are reliable and similar. The dynamic seasonally recurrent snow depletion pattern is only applicable during the melting period where other cloud removal algorithms like MOD10A1F ensure continuous cloud free imagery the whole water year (Hall et al. 2019). Snow cover estimates are most useful during the melting period because of their importance for calculation of sensible and latent heat fluxes at the snow-to-snow-free interface. Regional differences in drought may also produce different results than we observed in the Boise River Basin. We chose the Boise River Basin because of the high frequency of drought (Table 2) and the consequences of drought for stakeholders.

5 | Conclusions

We analysed the importance of drought for model development and CGF with the dynamic seasonally recurrent snow depletion pattern. In the Boise River Basin drought and maximum snow extent and depletion are not correlated; however, during moderate to severe drought the progression of snow depletion can have large spatial differences. The dynamic seasonally recurrent

pattern of snow depletion derived with as few as 3 FDL composites generates results similar to the model derived from 17 FDL composites at the watershed (98.7% correlated) and local snowline scales (r-squared of 0.995). We tested the snow cover estimate similarity of CGF with PCA_{17} and found it to be 96.73% and it only slightly declined during severe drought to 94.76%. The PCA_{17} is also on average 95.64% similar to the cloud free pixels given by the MOD10A1F CGF data; however, snow cover similarity percentage is lower overall for MOD10A1F. CGF with the dynamic seasonally recurrent snow depletion pattern is flexible with an accurate snowline representation even during severe drought. Accurate snowline representation increases snowmelt runoff modelling performance and water management (Bennett et al. 2019). Future research is needed in areas with varied climates and snow regimes to quantify CGF accuracies. Spatial snow depletion pattern-based CGF methods rapidly produce daily data that supplies stakeholders with the critical snow coverage data necessary to model and manage snowmelt dominated water resources in the western United States and similar systems worldwide.

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Data Availability Statement

All data presented in this study are openly available in National Snow and Ice Data Center (NSIDC) <http://nsidc.org/> and the US drought monitor (<http://drought.gov/>).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Correlations between FDL composites are given. The gradient of light orange to dark orange indicates increasing correlation values. FDL by year are given along the vertical and horizontal axes. **Table S1:** Factor loadings and variance described by principal components 1 and 2 are given. Factor loadings are the correlations of the principal component to the input data where the input data is and FDL for each year. Variance described by each principal component is calculated during the analysis and the sum of variance described by all principal components is one.